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S8-65-6A-141**PRODUCT GROUP**

Rev.1.0

ISSUE DATE**2016/3/7****SPECIFICATION
FOR
APPROVAL**

() Preliminary Specification

(●) Final Specification

Title

Customer Approval Sheet for Lenovo 6.95 HADS

SUPPLIER

BOE

MODEL

TV070WSM-TL1

Version

Ex. 1.0

SIGNATURE**DATE**

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Please return 1 copy for your confirmation with your signature and comments.

APPROVED BY**DATE**

秦磊

2016/3/7

REVIEWED BY

蔡斯特

2016/3/7

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S8-65-6A-141****PRODUCT GROUP****Rev.1.0****ISSUE DATE****2016/3/7****Contents**

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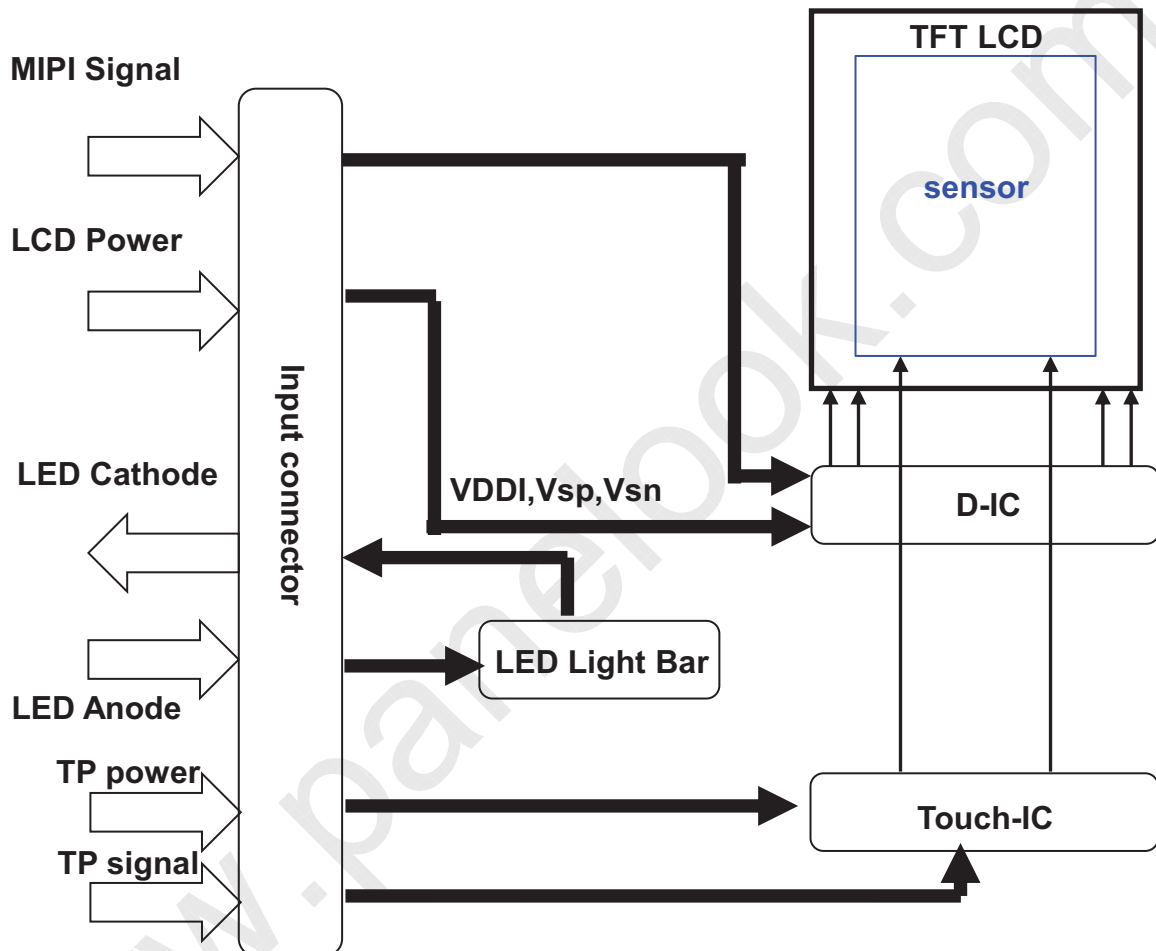
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1.0. GENERAL DESCRIPTION**Block Diagram****Features**

TV070WSM-TL1 is 7" color TFT-LCD (Thin Film Transistor Liquid Crystal Display) module composed of LCD panel, MIPI driver ICs, control circuit and backlight. By applying 8 bit digital data, $600 \times \text{RGB}(3) \times 1024$, 16.7M-color images are displayed on the 7" diagonal screen

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PRODUCT GROUP**Rev.1.0****ISSUE DATE****2016/3/7****1.1 General Specifications**

Parameter	Specification	Unit	Remarks
Screen Size	6.95	Inch	
Active Area	89.28*152.3712	mm	
Panel Size	93.28*159.82	mm	
Outline Dimension	95*163.27*2.4(LCM)	mm	(±0.15)
Display Resolution	600*RGB*1024	pixel	
Pixel Pitch	49.6*148.8	um	
Display Method	a-si	-	
Display Mode	Normal black	-	
Display Color	16.7M	-	
Color Gamut	typ:50%, min: 45%	%	NTSC
Luminance	typ: 300, min: 240	nit	Single Center Point Value
Contrast Ratio	typ 800:1,min 700:1	-	
Viewing Angle	85/85/85/85(CR>10)	-	Single Center Point
Pol Surface Treatment	HC	-	
Weight	Max: 125(TLCM)	g	
D-IC	HX8394D	-	
Inversion Method	Zig-Zag	-	
LED Q'ty	18	ea	
Power Consumption	350+1250	mw	Logic+Backlight

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ISSUE DATE**2016/3/7****1.2 Touch General Specifications**

item		Unit	Requirement	Remarks
TP Type		-	Signal Layer On-cell	
Cover meteraial		-	Sodalime 2nd	
sensor	Stack up	-	On-cell	
	driving	-	Multi-Touch	
Input method		-	Finger	
Fingers support		-	5-fingers	
Origin location		-	Up-Left	
Bonding Between TP And Display		-	Full lamination	
Glue (CG/L CM)	OCA/O CR	-	OCA	
	THK	mm	0.2	
Air Gap Between TP / Display		mm	0.2	
Thickness (CG+LCM)		mm	3.3 (± 0.15)	
AR Coating		-	No	
AF Coating		-	No	
AA Dimension		mm	89.28*152.3712	
VA Dimension		mm	90.08*153.17	
Sensor Outline Size (=CF)		mm	93.28*156.62	

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PRODUCT GROUP**Rev.1.0****ISSUE DATE****2016/3/7****1.3 Key Part List**

	item	Supplier	Spec/Size	Weight	Remark
Cell	TFT Glass	Dongxu	1850*1500*0.4T		mm
	C/F Glass	Dongxu	1850*1500*0.4T		mm
	Upper Pol	LGC	92.68*154.85*0.14mm	-	HC
	Lower Pol	LGC	92.68*154.85*0.124mm	-	Clear
	Liquid Crystal	Merck	BOE-F013	-	
	UV Glue	Sekisui	SWB-73	-	
FPCA	FPC	珠海元盛	63.55*25.7*0.25mm		
	EMI coating	NA			If necessary
	Main Connector	I-PEX	20613-039E-01		
	LED lightbar CNT	STM	MSAK24037P9		
	CNT Stiffener	珠海元盛	SUS		
	FPC Ass'y	珠海元盛			
BLU	Up Dif	升阳	PBS-689G 0.075T		
	Upper Prism	康得新	KBUO-100N 0.1T(135°)		
	Lower Prism	康得新	KBUO-100N 0.1T(45°)		
	Down Dif	LMS	DLAS-50D3 0.065T		
	LGP	三菱	HL7002(HL 8004) 0.57T		
	Mold frame	出光	URZ2501		
	Double Tape	德萨	Tesa4972 0.05T		
	Reflector	三菱	L20 150UM 0.15T		
	LED	聚飞	PS06W12 VO		
	LED Lightbar	SMT点极 / FPC 凯强	3806,18EA		
Total			-		

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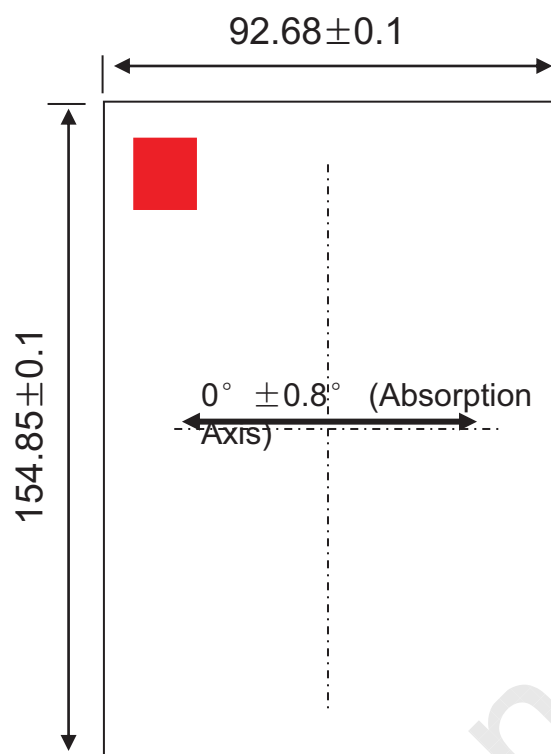
1.3.2. FPC Pin Assignment

Please pay attention that IC bump down(TFT glass up and C/F glass down)

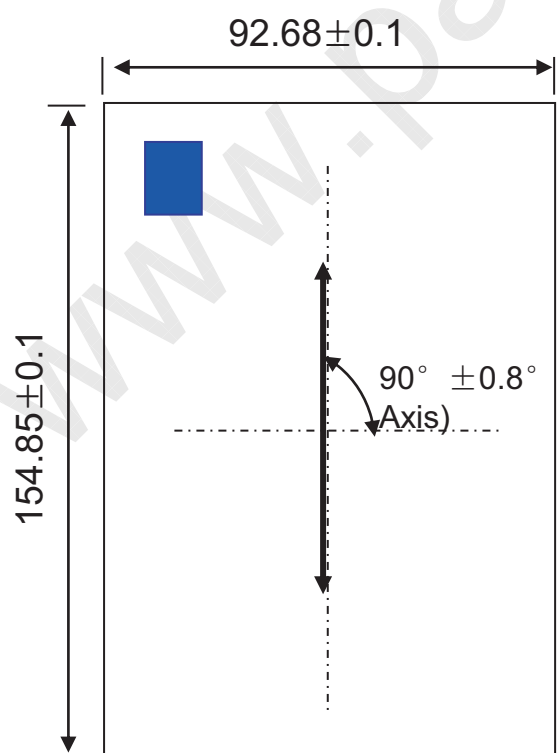
No.1	DUMMY	No.48	OPEN	No.94	VDD3
No.2	NULL	No.49	OPEN	No.95	OPEN
No.3	GND	No.50	OPEN	No.96	GND
No.4	GND	No.51	OPEN	No.97-98	GND(2)
No.5	VGOFF	No.52	OPEN	No.99	VSPR
No.6-7	VCOMOUT(2)	No.53	OPEN	No.100	VSNR
No.8	VGOFF	No.54	OPEN	No.101	VREF
No.19	GND	No.55	OPEN	No.102	GND
No.20	D1N	No.56	OPEN	No.103	VDD3
No.21	D1P	No.57	OPEN	No.104	VCL
No.22	GND	No.58	HSYNC	No.105	C41P
No.23	CLKN	No.59	VSYNC	No.106	C41N
No.9-10	VCOMOUT(2)	No.60	DE	No.107	OPEN
No.11-12	VCOMIN(2)	No.61	PCLK	No.108	OPEN
No.13-14	GND(2)	No.62	DCX	No.109	OPEN
No.15	OPEN	No.63	CSX	No.110	VSP
No.16	GND	No.64	SCL	No.111	VSP
No.17	D0N	No.65	SDI	No.112-113	VSN(2)
No.18	D0P	No.66	SDO	No.114	C21P
No.19	GND	No.67	PWMOUT	No.115	C21N
No.20	D1N	No.68	TE	No.116	C22P
No.21	D1P	No.69	TE1	No.117	C22P
No.22	GND	No.70	RESET	No.118	C22N
No.23	CLKN	No.71	GND	No.119	C22N
No.24	CLKP	No.72	GND	No.120	VGH
No.25	GND	No.73	GND	No.121	VDD3
No.26	D2N	No.74	BS0	No.122	VDD3
No.27	D2P	No.75	VDD1	No.123-124	GND(2)
No.28	GND	No.76	BS1	No.125	C31P
No.29	D3N	No.77	GND	No.126	C31N
No.30	D3P	No.78	Open	No.127	OPEN
No.31-32	GND(2)	No.79	VDD1	No.128	OPEN
No.33	HS_LDO	No.80	Open	No.129	OPEN
No.34	HS_VCC	No.81	GND	No.130	VGL
No.35	VDD1	No.82	BS2	No.131	VGL
No.36	VDDD	No.83	VDD1	No.132	OPEN
No.37	VDDD	No.84	PCCS0	No.133	OPEN
No.38-39	GND(2)	No.85	GND	No.134	VGL2
No.40	OPEN	No.86	PCCS1	No.135	OPEN
No.41	OPEN	No.87	VDD1	No.136-137	VCOMIN(2)
No.42	OPEN	No.88	VDDD	No.138-139	VCOMOUT(2)
No.43	OPEN	No.89	GND	No.140	VGOFF
No.44	OPEN	No.90	VCSW2	No.141-142	VCOMOUT(2)
No.45	OPEN	No.91	VCSW1	No.143-145	GND(3)
No.46	OPEN	No.92	VPP	No.146	NULL
No.47	OPEN	No.93	OPEN	No.147	DUMMY

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Adhesive Downwards

HC+Clear($140\mu\text{m} \pm 25\mu\text{m}$)

Adhesive Downwards

Clear ($124\mu\text{m} \pm 25\mu\text{m}$)

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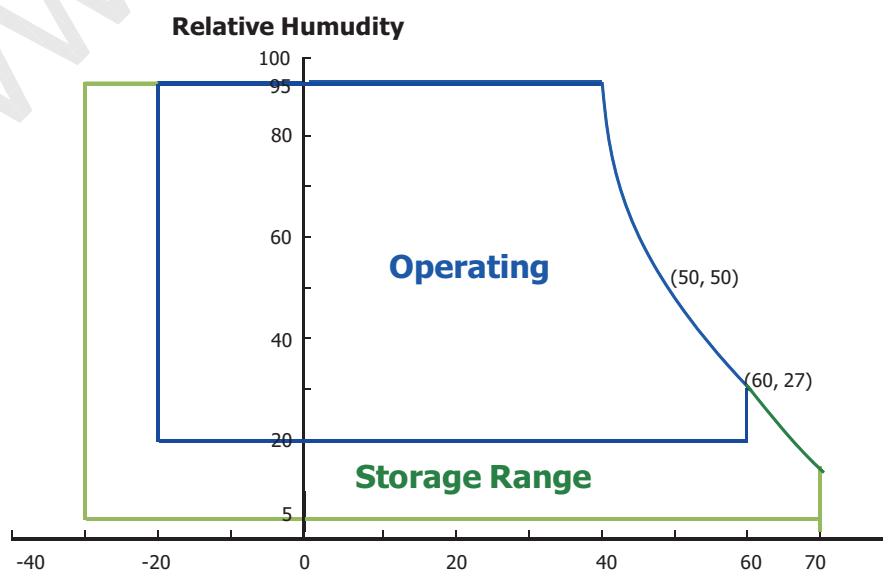
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The followings are maximum values which, if exceed, may cause faulty operation or damage the unit. The operational and non-operational maximum voltage and current values are listed in the following table .

Parameter	Symbol	Min.	Max.	Unit	Remarks
Logic Power Supply Voltage	V_{DD}	-0.3	3.6	Vdc	Note1
LED Forward Voltage of every LED string	V_{LED}	-	19.2	Vdc	
LED Forward Current of every LED string	I_{LED}	-	22	mA	Note2
LED string Reverse Voltage	V_R	-	2	V	
Operating Temperature	T_{OP}	-20	+60	°C	Note3
Storage Temperature	T_{ST}	-20	+70	°C	

- Notes : 1. Permanent damage to the device may occur if maximum values are exceeded functional operation should be restricted to the condition described under normal operating conditions.
2. the max value of LED forward current is relative to ambient temperature,the correlation is show in figure 1.
3. Temperature and relative humidity range are shown in the figure below.
- 95 % RH Max. (40 °C ≥ Ta)
- Maximum wet - bulb temperature at 39 OC or less. (Ta > 40 OC) No condensation.



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PRODUCT GROUP**Rev.1.0****ISSUE DATE****2016/3/7****3.0 Electrical Specifications**

[Ta =25±2 °C]

Parameter	Symbol	Values			Unit	Notes
		Min	Typ	Max		
Power Supply Input Voltage	VDD3V3	2.5	3.3	3.6	Vdc	1
Power Supply Ripple Voltage	VRP	-	300	360	mV	
Power Supply for TP	TP_VDD	2.8	3.3	3.6	Vdc	
Power Supply for TP	TP_IOVCC	1.75	1.8	1.85	Vdc	
LED Forward Voltage of every LED string	V _{LED}	-	18	19.2	Vdc	
LED Forward Current of every LED string	I _{LED}	-	20	-	mA	@60°C
Logic Power Consumption	Plogic	-	-	0.35	W	@R/G/B
BLU Power Consumption	Pblu	-	1.08	1.152	W	20mA
Rush current	IRUSH	-	0.7	1	A	2

Notes : 1. The supply voltage is measured and specified at the interface connector of LCM.

The current draw and power consumption specified is for VDD=3.3V, Frame rate fV=60Hz and Clock frequency = 42.4MHz. Test pattern of power supply current is : Typ. @White and Max. @Red Pattern

2. The duration of rush current is about 2ms and rising time of Power input is 1ms(min)

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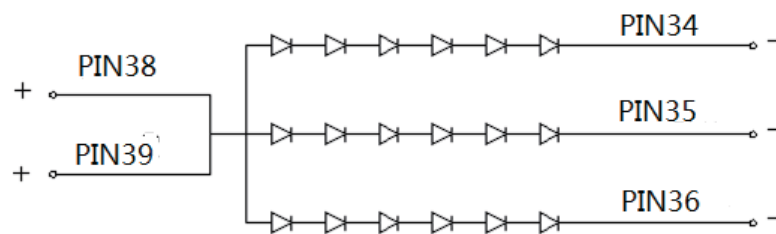
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3.1 BACK LIGHT UNIT

The edge-lighting type of back light unit consists of 18 LEDs which is connected in serial.

Table 3.1 Electrical Characteristics Of Back Light Unit

Parameter	Symbol	Values			Units	Notes
		Min	Typ.	Max		
LED Current	I_{LED}	-	20	-	mA	
LED Forward Voltage	V_{LED}	2.8	3.0	3.2	V	



LED Circuit Diagram

3-1-1 LED Rank

Luminance Rank : 8.0 ~ 8.25 lm

Color Rank : 2CF, 2CG, 2DF, 2DG, 2EF, 2EG

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PRODUCT GROUP**Rev.1.0****ISSUE DATE****2016/3/7****3.2. LCD INTERFACE CONNECTIONS**

Interface Connector: FPC Connector FH26W-39S-0.3SHW(06) is used for the module electronics interface.

<Table 3.2. Pin Assignments for the Interface Connector>

Terminal	Symbol	Functions
Pin No.	Symbol	Description
1	VLED	LED+
2	VLED	LED+
3	MTP	Power supply for OTP
4	FB1	LED-
5	FB2	LED-
6	FB3	LED-
7	NC	NC
8	TP_RST	Signal for TP
9	TP_ATTEN	Signal for TP
10	TP_SDA	Signal for TP
11	TP_SCL	Signal for TP
12	GND	Ground
13	TP_VDDH	Signal for TP
14	TP_VBUS	Signal for TP
15	GND	Ground
16	RESET	Device Reset Signal
17	LEDPWM	PWM output
18	NC	NC
19	NC	NC
20	NC	NC

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<Table 3.2. Pin Assignments for the Interface Connector>

Terminal	Symbol	Functions
Pin No.	Symbol	Description
21	GND	Ground
22	VDD3V3	Power Supply, 3.3V
23	VDD3V3	Power Supply, 3.3V
24	GND	Ground
25	D0N	MIPI Input Data Pair
26	D0P	MIPI Input Data Pair
27	GND	Ground
28	D1N	MIPI Input Data Pair
29	D1P	MIPI Input Data Pair
30	GND	Ground
31	CLKN	MIPI Input Clock Pair
32	CLKP	MIPI Input Clock Pair
33	GND	Ground
34	D2N	MIPI Input Data Pair
35	D2P	MIPI Input Data Pair
36	GND	Ground
37	D3N	MIPI Input Data Pair
38	D3P	MIPI Input Data Pair
39	GND	Ground

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4.0. SIGNAL TIMING SPECIFICATIONS

Signal	Item	Symbol	Min.	Typ.	Max.	Unit
	MIPI Data frequency	FDATA	145	150	200	MHz
DCLK	Frequency	1/Tc	-	42.4	-	MHz
DE	Vertical Total Time	TV	1040	-	-	TH
	Vertical Active Display Period	TVD	-	1024	-	TH
	Vertical Front Porch Period	TVFP	6	-	-	TH
	Vsync pulse width	TVPW	2	-	-	TH
	Vertical Back Porch Period	TVBP	8	-	-	TH
	Horizontal Total Time	TH	680	-	-	Tc
	Horizontal Active Display Period	THD	-	600	-	Tc
	Horizontal Front Porch Period	THFP	20	-	-	Tc
	Horizontal pulse width	THPW	24	-	-	Tc
	Horizontal Back Porch Period	THBP	36	-	-	Tc

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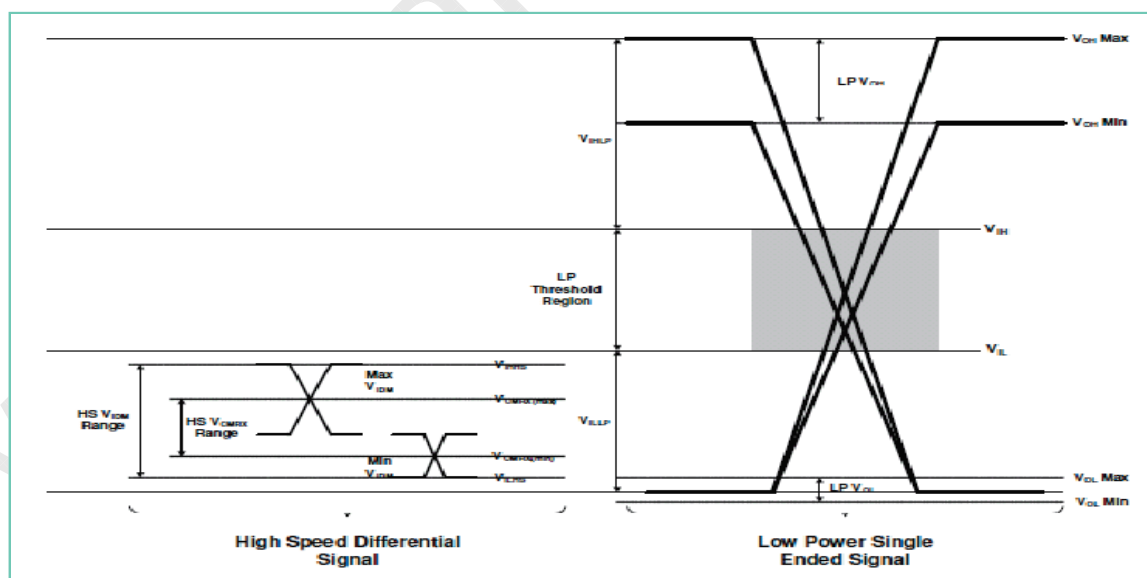
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4.1. MIPI Interface DC/AC Characteristic

MIPI Interface Timing Sequence

Parameter	Symbol	Min	Typ	Max	Unit	Condition
MIPI digital operation current	I_{VCCIF}	14	15	16	mA	-
MIPI digital stand-by current	$I_{VCCIFST}$	-	200	-	uA	-
MIPI Characteristics for High Speed Receiver						
Single-ended input low voltage	V_{ILHS}	-40	-	-		
Single-ended input high voltage	V_{IHHS}	-	-	460	mV	
Common-mode voltage	V_{CMRXDC}	155	-	330	mV	
Differential input impedance	Z_{ID}	80	100	125	Ω	
HS transmit differential voltage($V_{OD}=V_{DP}-V_{DN}$)	$ V_{OD} $	85	200	250	mV	
MIPI Characteristics for Low Power Receiver						
Pad signal voltage range	V_I	-50	-	1350	mV	
Ground shift	V_{GNDSH}	-50	-	50	mV	
Output low level	V_{OL}	-150	-	150	mV	
Output high level	V_{OH}	1.1	1.2	1.3	V	



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The Display Serial Interface standard defines protocols between a host processor and peripheral devices that adhere to MIPI Alliance standards for mobile device interfaces. The DSI standard builds on existing standards by adopting pixel formats and command set defined in MIPI Alliance standards.

DSI-compliant peripherals support either of two basic modes of operation: Command Mode and Video Mode. Which mode is used depends on the architecture and capabilities of the peripheral. The mode definitions reflect the primary intended use of DSI for display interconnect, but are not intended to restrict DSI from operating in other applications.

Command Mode refers to operation in which transactions primarily take the form of sending commands and data to a peripheral, such as a display module, that incorporates a display controller. The display controller may include local registers. Systems using Command Mode write to, and read from the registers. The host processor indirectly controls activity at the peripheral by sending commands, parameters and data to the display controller. The host processor can also read display module status information. Command Mode operation requires a bidirectional interface.

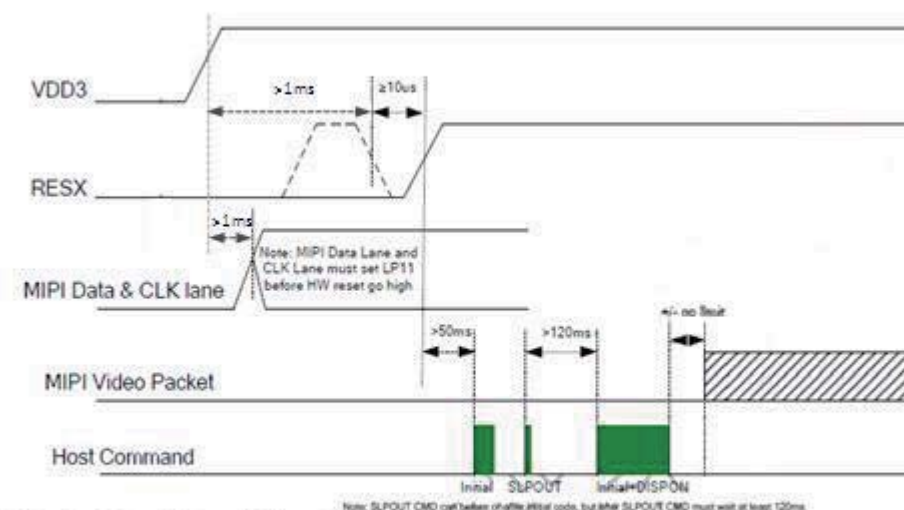
Video Mode refers to operation in which transfers from the host processor to the peripheral take the form of a real-time pixel stream. In normal operation, the display module relies on the host processor to provide image data at sufficient bandwidth to avoid flicker or other visible artifacts in the displayed image. Video information should only be transmitted using High Speed Mode. To reduce complexity and cost, systems that only operate in Video Mode may use a unidirectional data path.

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S8-65-6A-141**PRODUCT GROUP****Rev.1.0****ISSUE DATE****2016/3/7****4.2. Power On/Off Sequence** Based on HX8394D

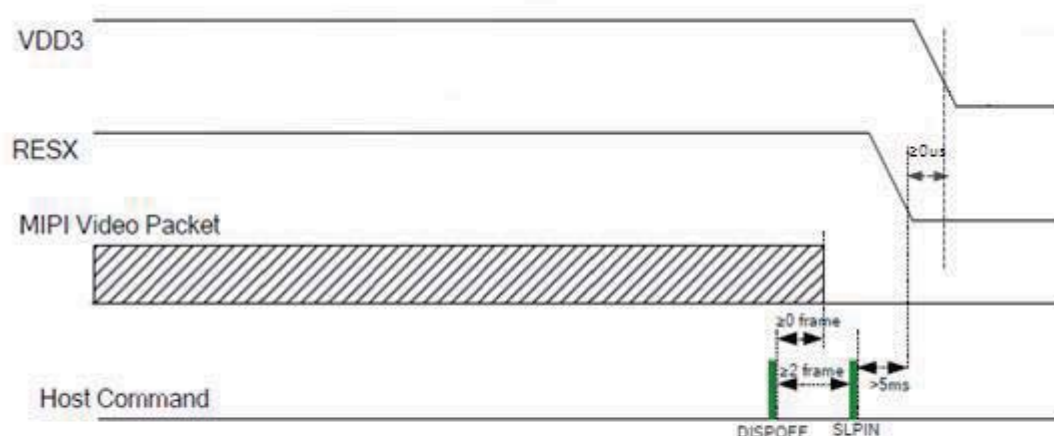
(1). Power on Timing Sequence:



Note: For External Power Mode not need VDD3

VDD3 input power on sequence

(2). Power off Timing Sequence:



Note: For External Power Mode not need VDD3

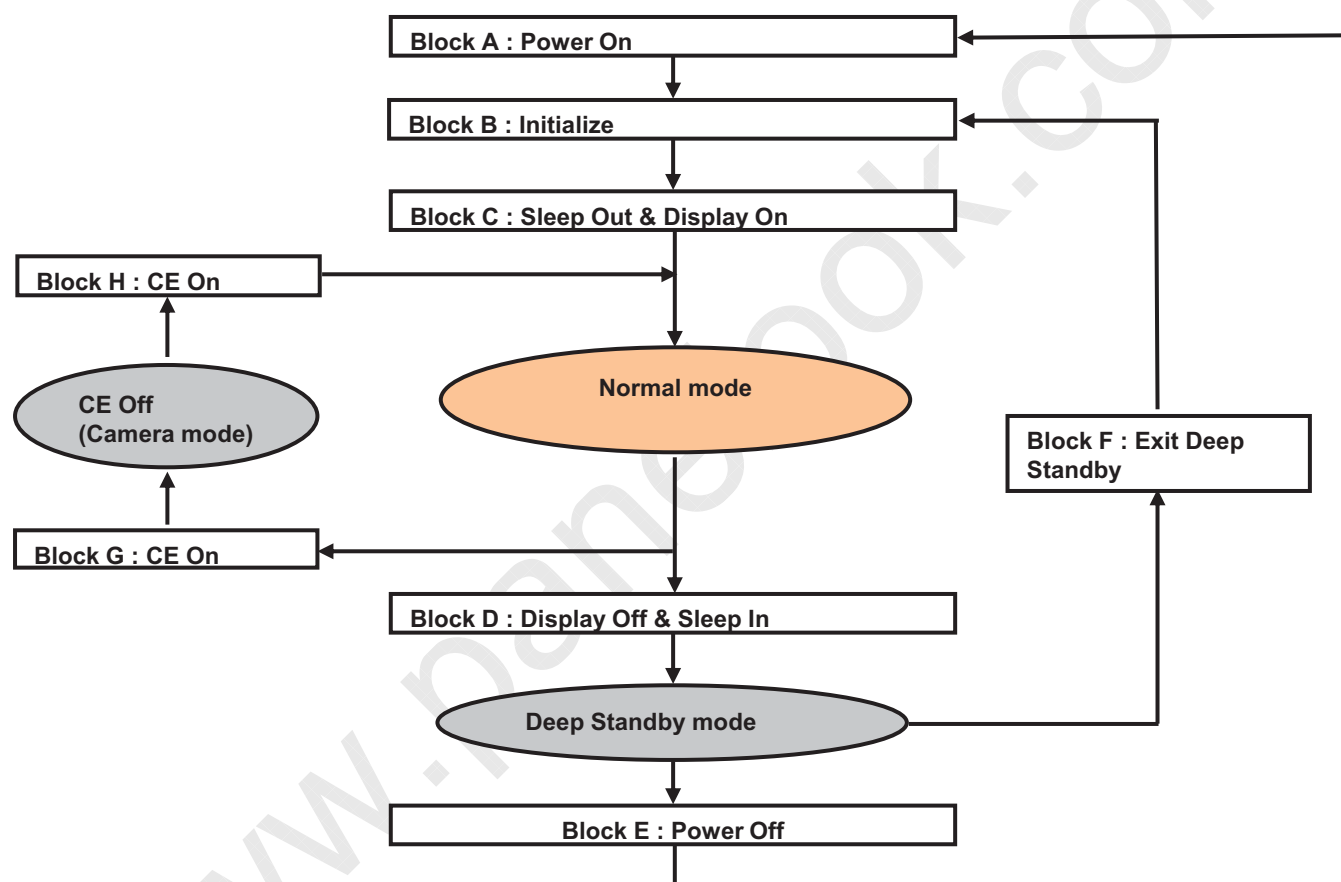
VDD3 input power off sequence

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4.4. IC General Spec and Size

FEATURES

Display

- Single chip solution for a WXGA type TFT LCD display
- Resolution:
 - 800RGB x (480 + (8 x NL))
 - 768RGB x (480 + (8 x NL))
 - 720RGB x (480 + (8 x NL))
 - 640RGB x (480 + (8 x NL))
 - 600RGB x (480 + (8 x NL))
- Display color modes
 - Full color mode:
 - 16.7M colours (24-bit 8(R):8(G):8(B))
 - Reduce color mode:
 - 262k colours (18-bit 6(R):6(G):6(B))
 - 65k colours (16-bit 5(R):6(G):5(B))
 - 8 colors (Idle mode on): 8 colors (3-bit binary mode)

Display module

- Support 2402 source channel outputs
- Gate driver control signals for GIP
- Supports 1-dot / 2-dot / 4-dot / 8-dot / Zig-Zag / Column inversion
- Output voltage level
 - Positive source output voltage level: VSPR to VSSA
 - Negative source output voltage level: VSNR to VSSA
 - Positive gate driver output voltage level: VGH to VSSA
 - Negative gate driver output voltage level: VGL to VSSA
 - VCOM= -4V ~ -0.3V, each step=10mV

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SPEC. NUMBER
S8-65-6A-141

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Rev.1.0

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2016/3/7

4.5. Power Consumption

	Parameter	Symbol	Typ.	Unit	Remark
CABC OFF	Logic Power	L0	127.8	mW	
		L32	126.8	mW	
		L64	128.9	mW	
		I96	127.5	mW	
		L127	125.6	mW	
		L160	127.4	mW	
		L192	128.2	mW	
		I224	127.1	mW	
		L255	128.8	mW	
		R255	204.5	mW	
		G255	205.1	mW	
		B255	205.8	mW	
		8 color bar	187.4	mW	
	BLU	25% on	0.31	W	
		50% on	0.65	W	
		75% on	0.87	W	
		100% on	1.19	W	

	Symbol	Typ.	Unit	Remark
CABC On	L0	TBD	TBD	
	L32	TBD	TBD	
	L64	TBD	TBD	
	I96	TBD	TBD	
	L127	TBD	TBD	
	L160	TBD	TBD	
	L192	TBD	TBD	
	I224	TBD	TBD	
	L255	TBD	TBD	
	R255	TBD	TBD	
	G255	TBD	TBD	
	b255	TBD	TBD	
	8 color bar	TBD	TBD	

If there is more than one mode for CABC function, please fill out all data

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PRODUCT GROUP**Rev.1.0****ISSUE DATE****2016/3/7****4.6. Touch Performance**

item	unit	requirement	remark
Number of touch	point	5-finger	
Ito pitch	mm	6.2*7.0	
Accuracy	mm	Edge area<2.5 Center area<1.5	
Presision	mm	Edge area<2.5 Center area<1.5	
Linearity	mm	Edge area<2.5 Center area<1.5	
Jitter	mm	Edge area<1.5 Center area<1.0	
Response time	ms	50	
Finger	mm	5	
Sensitivity	mm	Φ6	
Separation	mm	15	
Report rate	Hz	80	
Latency	-	/	
Signal-to-noise raito	-	30	
Passive Pen Support	-	NO	
Glove input	-	NO	
Anti-Water	-	YES	
Hover	-	NO	
Point report interval	-	/	

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S8-65-6A-141****PRODUCT GROUP****Rev.1.0****ISSUE DATE****2016/3/7****4.6.1 Cover Glass**

item		unit	requirement	remark
Thickness		mm	0.7	
Material		-	Sodalime 2nd	
Hardness		H	≥7H	
intensity	DOL	um	10~12	
	CS	MPa	530~600	
Main printing color		-	黑	
Logo printing color		-	-	无Logo

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5.0 Optical Specifications -Backlight 100%

The test of Optical specifications shall be measured in a dark room (ambient luminance ≤ 1 lux and temperature = $25 \pm 2^\circ\text{C}$) with the equipment of Luminance meter system (CA-310、BM-5A) and test unit shall be located at an approximate distance 50cm from the LCD surface at a viewing angle of θ and Φ equal to 0° .

Parameter		Symbol	Condition	Min.	Typ.	Max.	Unit	Remark
Viewing Angle range	Horizontal	Θ_3	CR > 10	80	85	89	Deg.	Note 1
		Θ_9		80	85	89	Deg.	
	Vertical	Θ_{12}		80	85	89	Deg.	
		Θ_6		80	85	89	Deg.	
Color Gamut				45	50	55	%	-
Luminance Contrast ratio		CR	$\Theta = 0^\circ$	700:1	800:1	1000:1	-	Note 2
Luminance of White	Center Point	Y_w	$\Theta = 0^\circ$	240	300	-	cd/m ²	Note 3 (TLCM)
White Luminance uniformity	13 Points	ΔY_{13}		70	-	-	%	Note 4
	5 Points	ΔY_5		75	-	-	%	Note 4
White balance		W_x	$\Theta = 0^\circ$	0.27	0.30	0.33	-	Note 5
		W_y		0.29	0.32	0.35	-	
Reproduction of color	Red	R_x	$\Theta = 0^\circ$	0.558	0.588	0.618	-	Note6 (TLCM)
		R_y		0.309	0.339	0.369		
	Green	G_x		0.275	0.305	0.335		
		G_y		0.545	0.575	0.605		
	Blue	B_x		0.122	0.152	0.182		
		B_y		0.091	0.121	0.151		
Response Time (Rising + Falling)		T_{RT}	Ta= 25° C $\Theta = 0^\circ$	25	30	35	ms	Note 7

Cell & BLU Optical Characteristics

Parameter	Typ	Unit	Remarks
Aperture Ratio	70	%	
Upper Pol Trans.	43	%	
Lower Pol Trans.	43	%	
Panel Trans.	4.6(TLCM)	%	w/o APF
Panel Trans.	-	%	with APF
BLU Luminance	6500	Cd/m ²	Center

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Note :

1. Viewing angle is the angle at which the contrast ratio is greater than 10. The viewing angles are determined for the horizontal or 3, 9 o'clock direction and the vertical or 6, 12 o'clock direction with respect to the optical axis which is normal to the LCD surface (see FIGURE 1).
2. Contrast measurements shall be made at viewing angle of $\Theta = 0$ and at the center of the LCD surface. Luminance shall be measured with all pixels in the view field set first to white, then to the dark (black) state. (see FIGURE 1)

1) Luminance Contrast Ratio (CR) is defined mathematically.

$$CR = \frac{\text{Luminance when displaying a white raster}}{\text{Luminance when displaying a black raster}}$$

3. Center Luminance of white is defined as luminance values of 1point average across the LCD surface. Luminance shall be measured with all pixels in the view field set first to white. This measurement shall be taken at the locations shown in FIGURE 1 for a total of the measurements per display. The luminance is measured by CA310 when the LED current is set at 20mA.
4. The White luminance uniformity on LCD surface is then expressed as : $\Delta Y = \text{Minimum Luminance of 9points} / \text{Maximum Luminance of 9points}$ (see FIGURE 2).
5. The color chromaticity coordinates specified shall be calculated from the spectral data measured with all pixels first in red, green, blue and white. Measurements shall be made at the center of the panel.
6. The color chromaticity coordinates specified shall be calculated from the spectral data measured with all pixels first in red, green, blue and white. Measurements shall be made at the center of the panel.
7. The electro-optical response time measurements shall be made as FIGURE 3 by switching the "data" input signal ON and OFF. The times needed for the luminance to change from 10% to 90% is T_r , and 90% to 10% is T_d .

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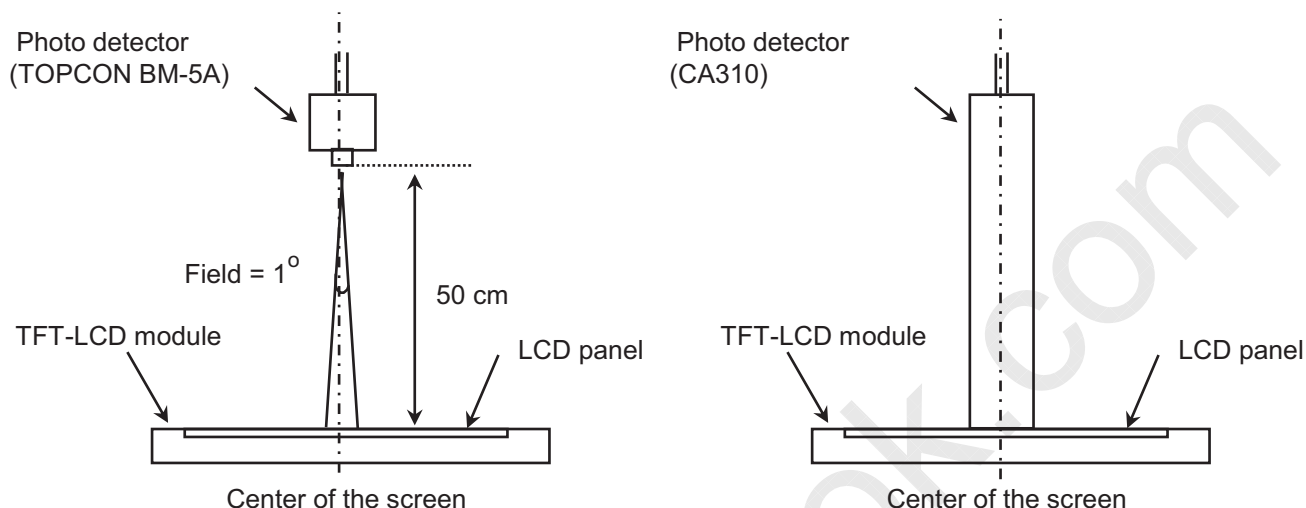
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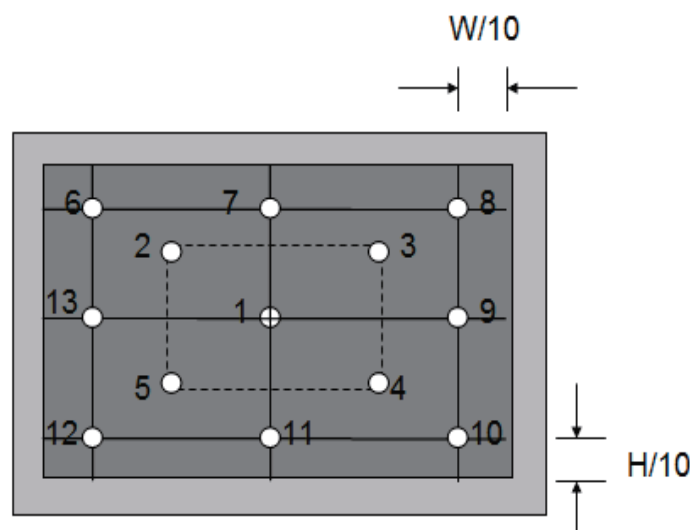
2016/3/7

Figure 1. Measurement Set Up



View angle range measurement setup Luminance , uniformity and color measurement setup

Figure 2. White Luminance and Uniformity Measurement Locations (13 points)



Center Luminance of white is defined as luminance values of center 9 points across the LCD surface. Luminance shall be measured with all pixels in the view field set first to white. This measurement shall be taken at the locations shown in FIGURE 2 for a total of the measurements per display.

$$\text{Brightness Uniformity} = \frac{\text{Minimum Photo detector output for P1-P13 with all pixels white}}{\text{Maximum Photo detector output for P1-P13 with all pixels white}}$$

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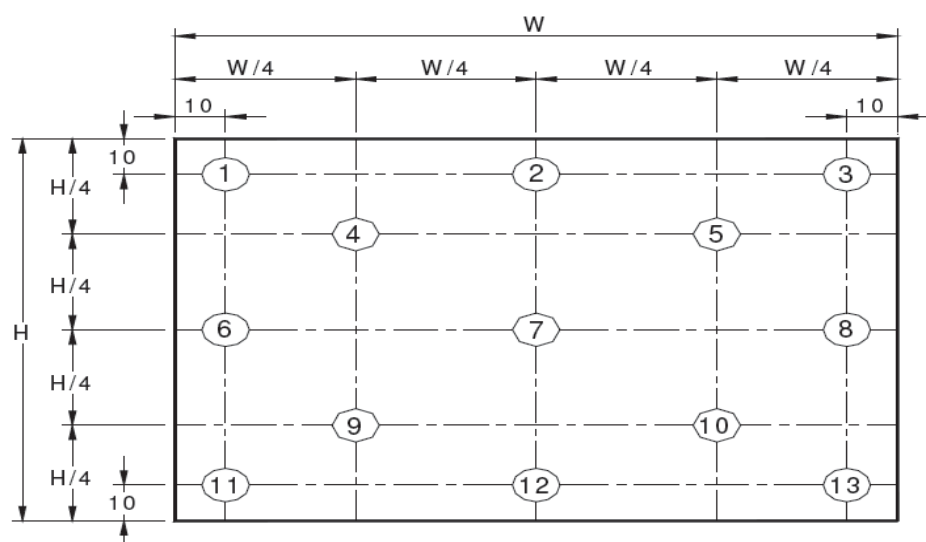
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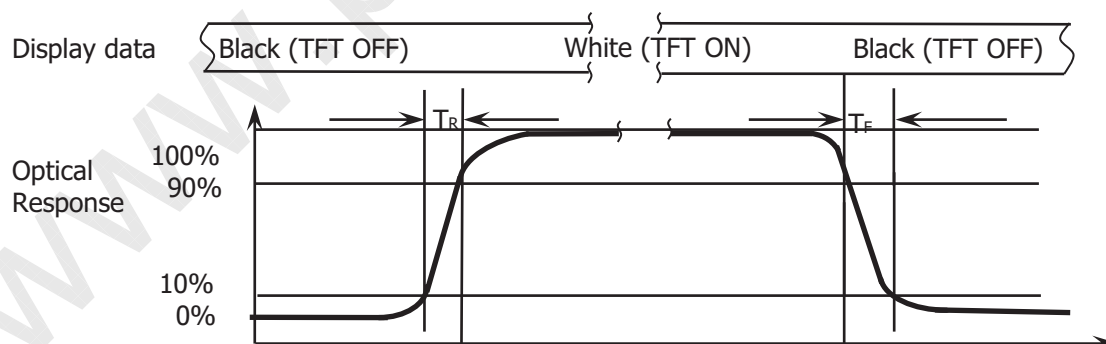
Figure 3. Uniformity Measurement Locations (13 points)



The White luminance uniformity on LCD surface is then expressed as : $\Delta Y_{13} = \text{Minimum Luminance of 13 points} / \text{Maximum Luminance of 13 points}$ (see FIGURE 3).

The White luminance uniformity of 5 point is the same test method as 13 point using FIGURE 4.

Figure 4. Response Time Testing



The electro-optical response time measurements shall be made as shown in FIGURE 4 by switching the "data" input signal ON and OFF. The times needed for the luminance to change from 10% to 90% is T_r and 90% to 10% is T_d .

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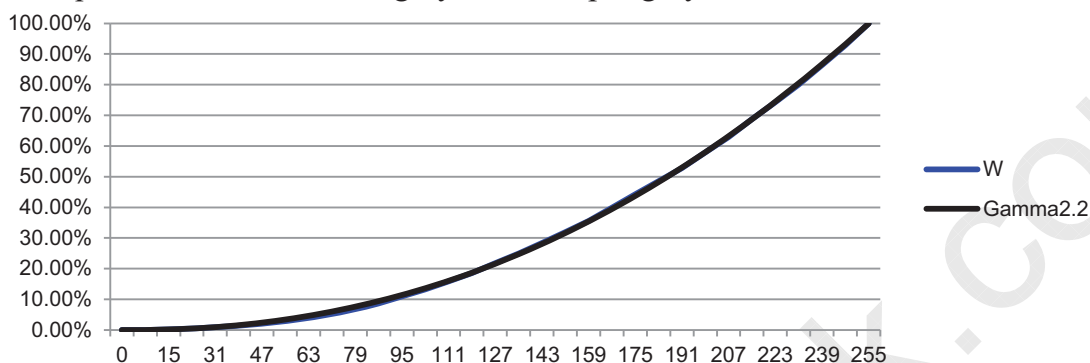
ISSUE DATE

2016/3/7

5.1. Gamma/Color Coordinate Uniformity

(1) Gamma Curve

Request: R/G/B/W, 0-255 gray scale, step 8 gray scale



(2) Color Coordinate Uniformity

Request: white pattern, 0-255 gray scale, step 8 gray scale.

Gary scale	x	y	Gary scale	x	y
0	0.2679	0.264	135	0.2999	0.32
7	0.2766	0.2752	143	0.3002	0.3206
15	0.292	0.2988	151	0.3004	0.3209
23	0.2999	0.3119	159	0.3008	0.3215
31	0.2955	0.3079	167	0.3005	0.3212
39	0.2934	0.306	175	0.3004	0.3209
47	0.291	0.3037	183	0.3005	0.3209
55	0.2911	0.3046	191	0.3005	0.321
63	0.2923	0.3068	199	0.3006	0.3212
71	0.2933	0.3085	207	0.3008	0.3213
79	0.2946	0.3108	215	0.301	0.3214
87	0.2959	0.3128	223	0.3012	0.3217
95	0.2973	0.3153	231	0.3012	0.3215
103	0.2974	0.3159	239	0.3012	0.3213
111	0.298	0.3168	247	0.3009	0.3208
119	0.2988	0.3181	255	0.3007	0.3201
127	0.2997	0.3198			

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S8-65-6A-141****PRODUCT GROUP****Rev.1.0****ISSUE DATE****2016/3/7****6. MECHANICAL CHARACTERISTICS**

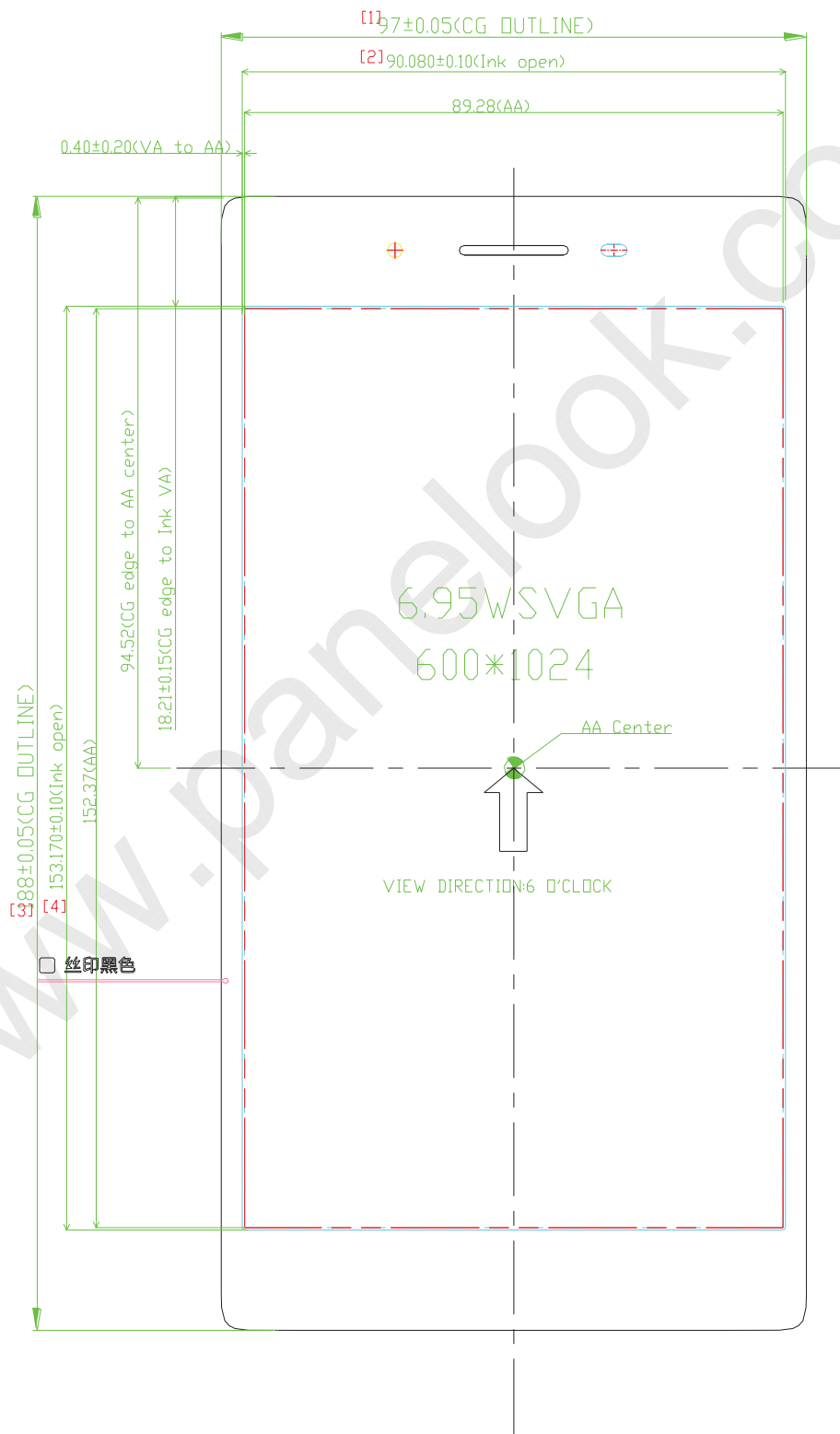
The contents provide general mechanical characteristics for the model.

In addition the figures in the next page are detailed mechanical drawing of the LCD.

item		Description	Typ.	Tolerance	Unit
Mother glass		Size	1850*1500	-	mm
CF and TFT thickness after slimming		thickness	0.4/ 0.4		mm
Panel	AA	A/A	89.28*152.371	-	mm
	CF	C/F	93.28*156.62	±0.2	mm
	TFT	TFT	93.28*159.82	±0.2	mm
	BM	BM(U/D/L/R)	1.83/5.62 /2.0/2.0	-	mm
	IC Bonding area	IC Bonding Area	3.2	-	mm
	Pol size	Pol Size	CF: 92.68*154.8 5	±0.2	mm
	Gap between pol~glass(U/D/L/R)	Gap Between Pol~C/F border (U/D/L/R)	CF: 0.3/1.47/0.3 /0.3	±0.2	mm
Module	Horizontal	Horizontal	97	±0.05	mm
	Vertical	Vertical	188	±0.05	mm
	Thickness	Thickness	2.4	±0.15	mm
	Uv glue thickness	UV Glue Thickness	-	-	mm

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SPEC. NUMBER
S8-65-6A-141**PRODUCT GROUP****Rev.1.0****ISSUE DATE****2016/3/7****6.1 Touch Display Module Drawing****(1) Front side**

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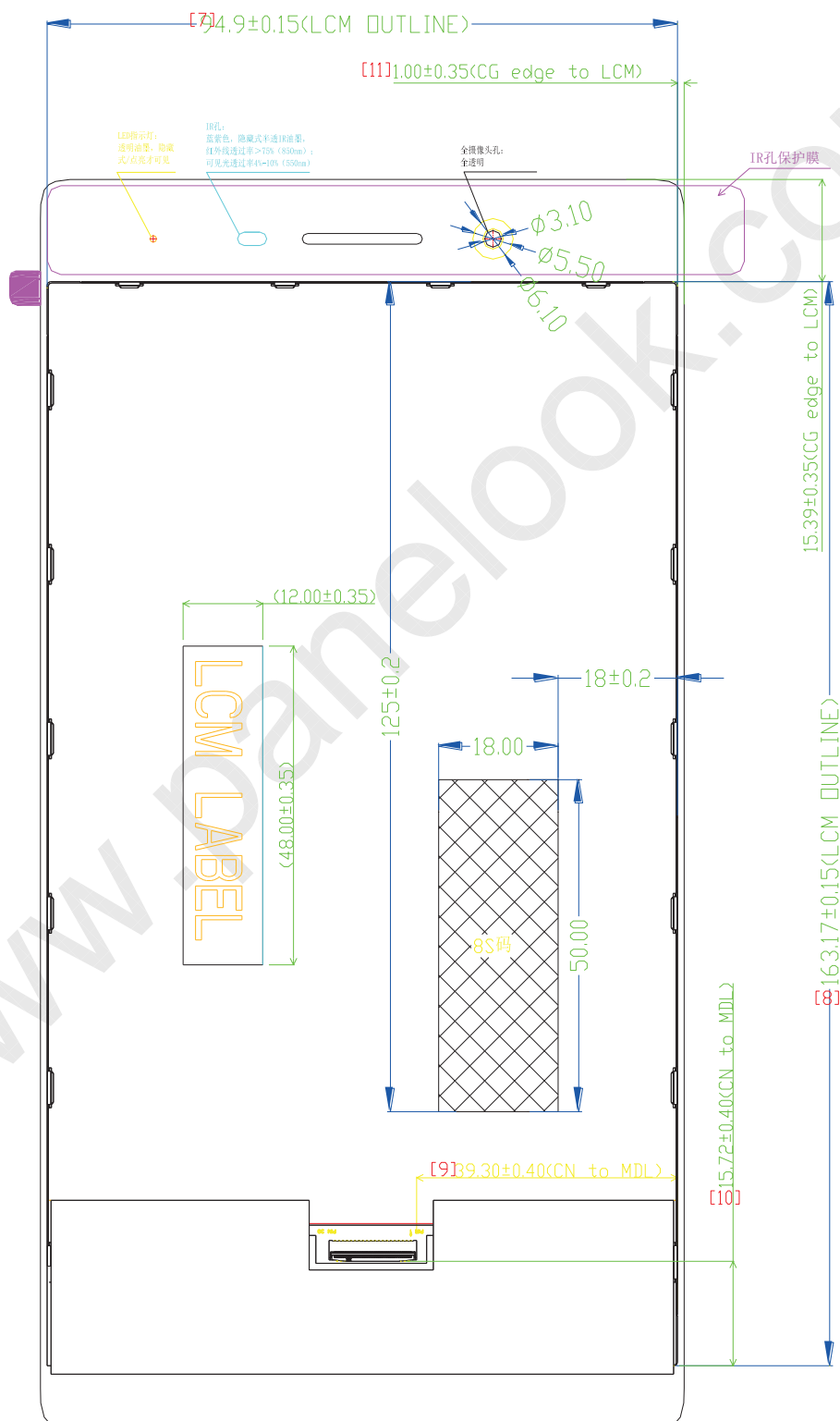
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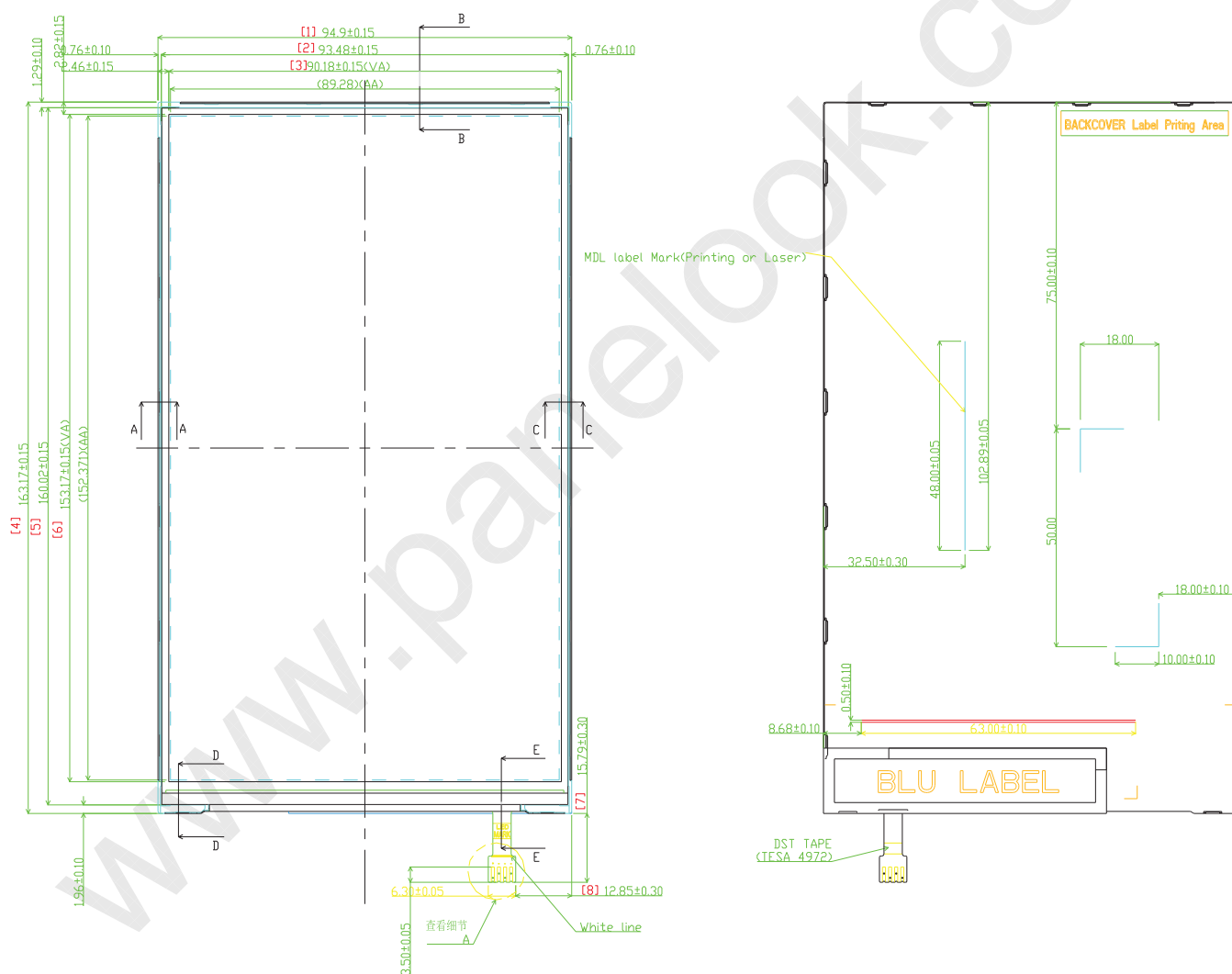
6.1 Touch Display Module Drawing

(2) Rear side



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S8-65-6A-141**PRODUCT GROUP****Rev.1.0****ISSUE DATE****2016/3/7****6.2 BLU Outline Dimension**

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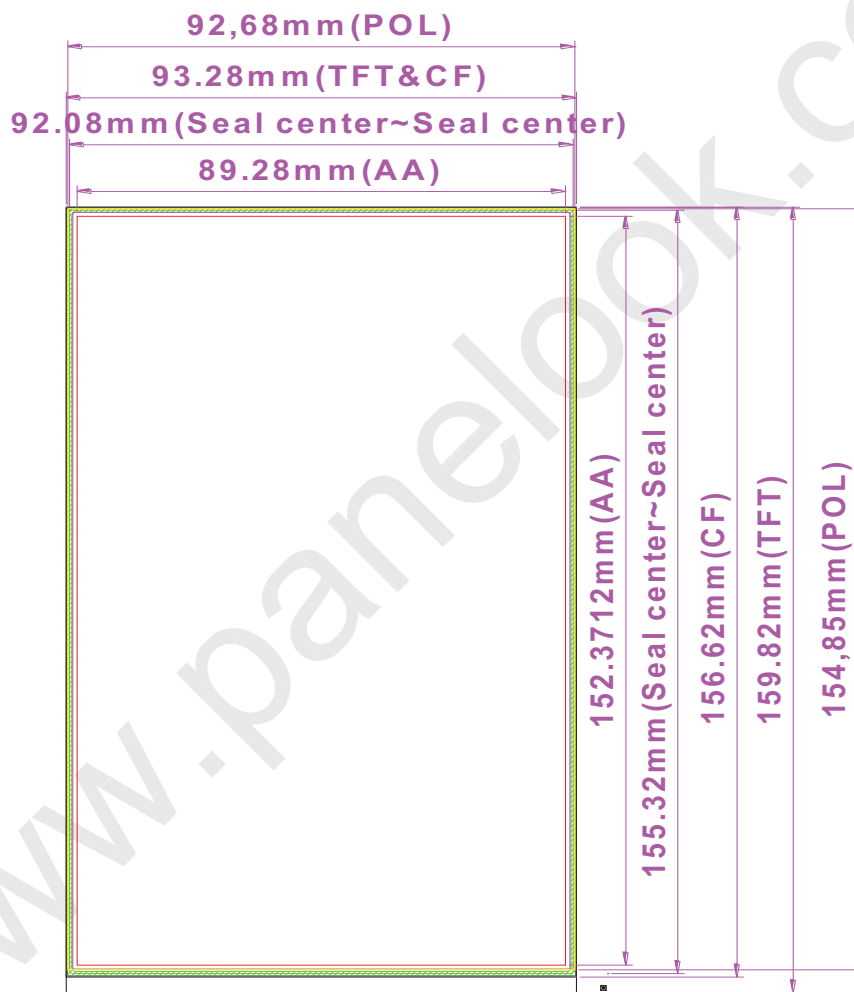
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6.3 Panel Outline Dimension



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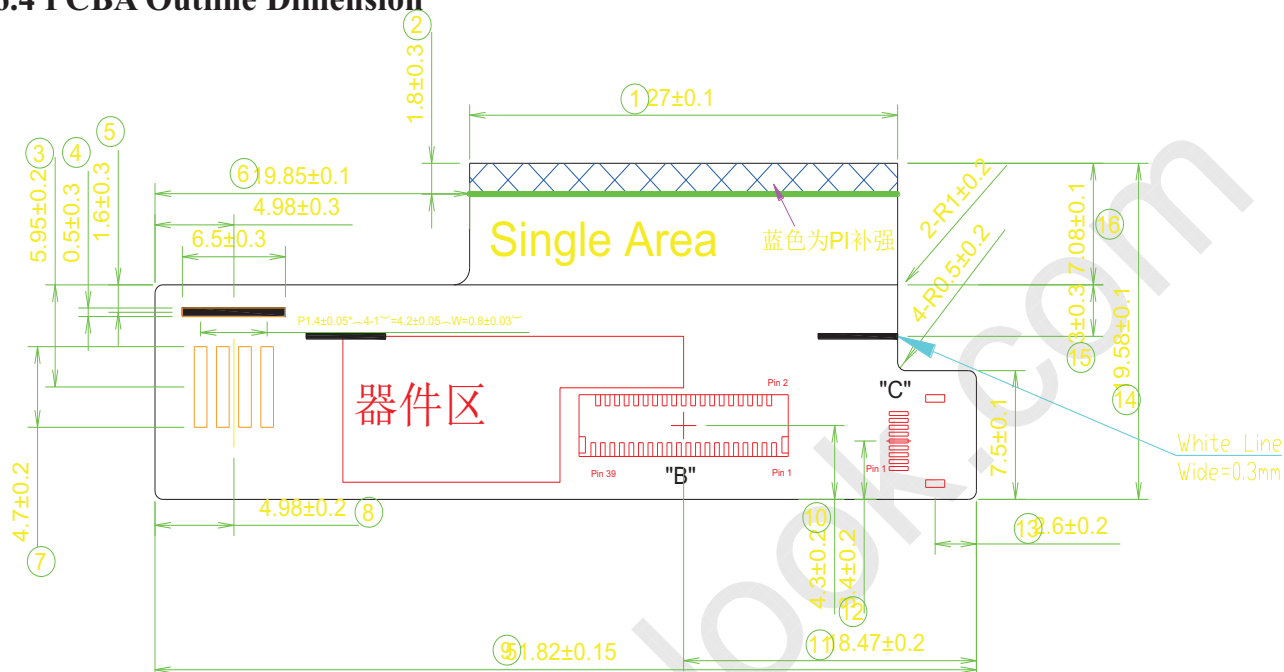
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6.4 PCBA Outline Dimension

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PRODUCT GROUP**Rev.1.0****ISSUE DATE****2016/3/7****7.0 Reliability Test**

No	Test Item	Test Condition	Remark
1	High temperature storage test	Ta = 70 °C , 240 hrs	-
2	Low temperature storage test	Ta = -20℃ , 240 hrs	
3	Thermal Shock Test	-20℃/0.5hr → 70℃/0.5hr × 100Cycle	
4	High temperature Operate test	Ta = 60 °C ,240hrs, 90%RH	
5	Low temperature Operate test	Ta = -20 °C , 240hrs	
6	High temperature High humidity	Ta = 60 °C , 96hrs	

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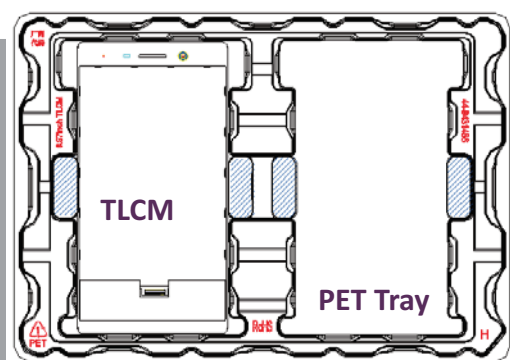
Rev.1.0

ISSUE DATE

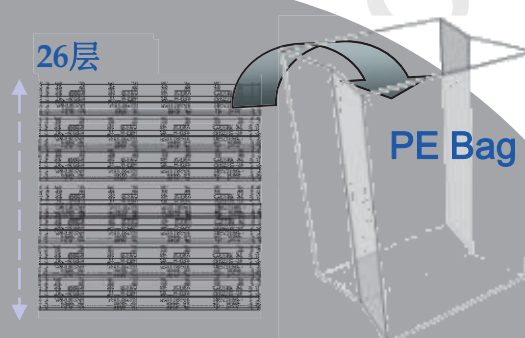
2016/3/7

8.0. Package

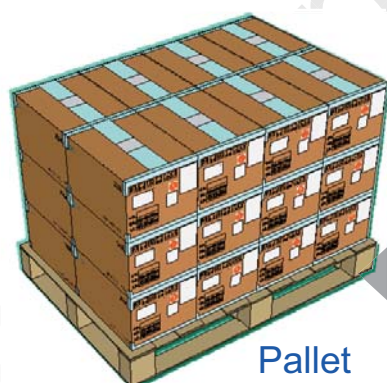
- 将 2pcs TLMC CG向下平放入Tray
- 产品上下放置EPE Spacer



- 将26pcs PET Tray 平放入PE Bag;
- Tray 无需旋转放置, 顶部1pcs 空Tray;



- 每个Pallet上放3层Box
- 1层8箱,共计24ea Box
- Pallet外进行缠膜包装
- 容量: 1200pcs/Pallet



Pallet

- 将26pcs PET Tray 平放入PE Bag;
- Tray 无需旋转放置, 顶部1pcs 空Tray;



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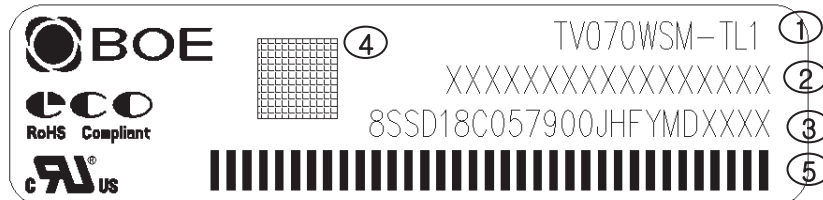
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2016/3/7

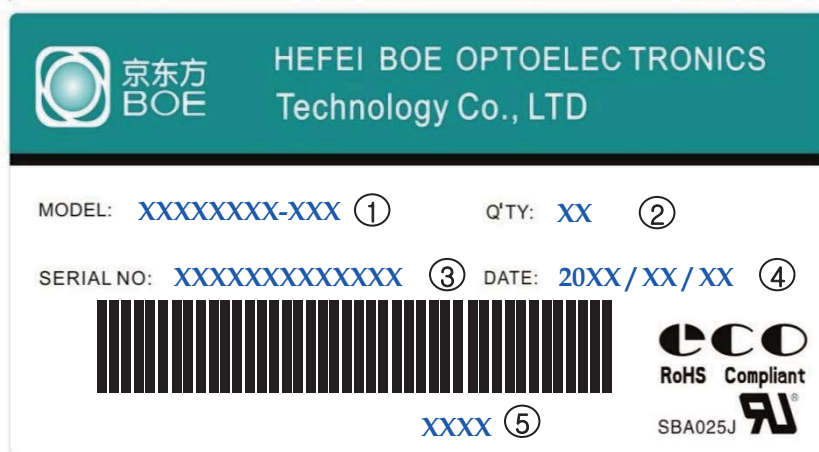
8.1 Box label



Remark:

标签贴附于背板Mark内, 标签尺寸: 48mm × 12mm × 0.08mm

1. FG-CODE: TV070WSM-TL1
2. MDL ID
3. 8S码料号: SD18C05790
4. 8S 码对应二维码
5. MDL ID 对应条形码



蓝色字体为后打印标识, 说明如下:

Label Size: 115mm*55mm

- 1.FG-CODE
- 2.Box 产品数量
- 3.Box ID, 编码规则如下
- 4.Box Packing 日期
- 5.FG-CODE 后四位

序列号	1	2	3	4	5	6	7	8	9	10	11	12	13
代码	X	X	P	3	1	5	7	0	0	0	1	H	D
描述	GBN代码		等级	B3	年份		月	Rev	序列号				

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9.0 Handing & Cautions

(1) Cautions when taking out the module

- Pick the pouch only, when taking out module from a shipping package.

(2) Cautions for handling the module

- As the electrostatic discharges may break the LCD module, handle the LCD module with care. Peel a protection sheet off from the LCD panel surface as slowly as possible.
- As the LCD panel and back - light element are made from fragile glass material, impulse and pressure to the LCD module should be avoided.
- As the surface of the polarizer is very soft and easily scratched, use a soft dry cloth without chemicals for cleaning.
- Do not pull the interface connector in or out while the LCD module is operating.
- Put the module display side down on a flat horizontal plane.
- Handle connectors and cables with care.

(3) Cautions for the operation

- When the module is operating, do not lose CLK, ENAB signals. If any one of these signals is lost, the LCD panel would be damaged.
- Obey the supply voltage sequence. If wrong sequence is applied, the module would be damaged.

(4) Cautions for the atmosphere

- Dew drop atmosphere should be avoided.
- Do not store and/or operate the LCD module in a high temperature and/or humidity atmosphere. Storage in an electro-conductive polymer packing pouch and under relatively low temperature atmosphere is recommended.

(5) Cautions for the module characteristics

- Do not apply fixed pattern data signal to the LCD module at product aging.
- Applying fixed pattern for a long time may cause image sticking.

(6) Other cautions

- Do not disassemble and/or re-assemble LCD module.
- Do not re-adjust variable resistor or switch etc.
- When returning the module for repair or etc., Please pack the module not to be broken. We recommend to use the original shipping packages.

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PRODUCT GROUP**Rev.1.0****ISSUE DATE****2016/3/7****10.0. Checklist**

	Description	Request	Value
Connection	MIPI Impedance	100±10Ω	Yes
	B2B CNT	ZIF	Stiffening:0.1T SUS304
	ID Pin	1 st source: GND 2 nd source: 1.8V(different cell) 2 nd source: GND(same cell+IC)	GND
IC	Protection	Shielding tape on IC	NO,if necessary
FPC	Bending Area	Not exceed M/F	No
	FPC status	Unfolded while direct bonding	Yes
	GND Area	Need GND area to connect SUS	Yes
	Test Points	Need shieling tape on it	PET tape
	Bending Area	Single layer	Yes
SUS	4 Corner side	At lease 1.2mm	No
Panel	Glass Generation	G5?G6?	G6
	Cutting Q'ty	panel Q'ty each mother glass	162
	Display mode	VA/IPS	ADS
	Mask Q'ty	Array mask	6
	Pixel Domain	1 or 2?	2
	ITO@C/F	ITO square resistance, thickness	38±8 1350A
	Scan direction	Single scan or dual scan	Single scan
	PS Parameter	Main/Sub PS density and size	Main : 1/21 Sub : 19/21 Size : (main&sub) 11/14
	Cell gap	Center point	3.35
	LC injection	Vacuum injection or ODF	ODF
	LC Margin		5%
	Pol compensation	A+C/B+B/None compensation film	None
	UV Glue	Fill out at IC around	Yes
	Pol surf. Treatment	Direct bonding: HC+Glare Air bonding: Haze44+glare(>4inch)	HC
	Pol position	Direct bonding: pol is higher 0.05mm than M/F Air bonding: pol is lower 0.05mm than M/F	Direct bonding: pol is higher 0.05mm than M/F
Package	Surface resistance	10 ⁴ ~10 ⁹ Ω	Yes
	Friction voltage	≦100V	Yes
	Layer in one Box	<10layer	